Contents

CHAPTER 1

1.0 Introduction

Health information management is central to effective healthcare delivery, as it supports patient care, reporting, and evidence-based planning. In maternity services, accurate birth records are particularly important because they provide legal proof of birth and essential data for monitoring maternal and child health outcomes, including live births and stillbirths. Despite global advances in digital and digitalised health record systems, Nyanga District Hospital continues to rely on handwritten birth registers. These paper-based records are vulnerable to damage, loss, illegibility, and slow retrieval, which negatively affects efficiency, data quality, and service delivery. This study focuses on the development of a Digitalised Birth Record Management System for Nyanga District Hospital. The proposed system aims to digitise birth records, improve accuracy through automated validation, and provide a dashboard that presents real-time summaries and trends. The dashboard will enable healthcare workers and management to easily view key indicators such as monthly birth rates, live births, stillbirths, delivery outcomes, and workload trends, thereby supporting faster reporting and informed decision-making. By integrating digital records with intelligent analytics and visualisation, the study seeks to strengthen birth record management and enhance maternity care services.

1.1 Background of the Study

At Nyanga District Hospital, birth registration and record management are carried out using physical handwritten birth record registers in the maternity ward and the Health Information Office. Whenever there is a need to verify or retrieve birth information, healthcare workers physically search through the registers to locate the required records. The records stored are from 1968 to the present.

Information relating to mothers and newborn babies is recorded in physical books and registers. Some newborn babies are issued birth records while others are not, depending on the mother’s decision or circumstances at the time of delivery. When a register becomes full, it is transferred to the Health Information Office for storage. However, the registers are kept without a proper filing order, making it difficult to trace old records when needed.

The physical system also faces challenges such as torn pages, faded ink, unclear handwriting, misplaced registers, and missing information. In cases where a record is missing, it is usually left unresolved or a letter is written to the Registry Department to request the missing information. These challenges affect the accuracy, accessibility, and management of birth records at the hospital

1.2 Statement of the Problem

The current system failed to overcome the following

- Birth records are recorded using handwritten registers, which are prone to torn pages, faded ink, unclear handwriting, and missing information.
- Paper-based registers are easily damaged or lost, putting important birth information at risk.
- Healthcare workers spend excessive time writing and managing paperwork instead of focusing on patient care.
- Retrieving specific birth records is slow and inefficient due to the accumulation of records over many years.

1.3 Objectives of the Study

1. To design and implement Digitalised Birth Record Management System.
2. To provide a dashboard that displays real-time reports and trends.

1.4 Hypotheses

H_0: The proposed system would solve the problem of issuing birth records at Nyanga District Hospital by 85%

H_1: The proposed system would not solve the problem of issuing birth records at Nyanga District Hospital.

1.5 Justification of the Research

This study is justified because effective birth record management is essential to healthcare delivery. Previous studies have largely focused on general electronic health records, overlooking digitalised birth record systems in district hospitals. Improving birth record management benefits healthcare workers, hospital management, parents, and policymakers. The study also contributes academically by addressing a neglected research area and practically by proposing a usable solution.

1.6 Assumptions of the Study

The study assumes that participants provide honest response, that basic infrastructure is available, and that healthcare workers are willing to adopt new technologies.

1.7 Delimitations of the Study

The study is limited to birth record management at Nyanga District Hospital. It focused on maternity ward records and healthcare workers involved in birth registration. Other hospital departments and national systems are excluded.

1.8 Limitations of the Study

Limitations include time constraints, limited access to older records, and varying levels of computer literacy. These are addressed through focused data collection and a simple system design.

1.9 Definition of Terms

- **Digitalised Birth Record Management System:** An electronic system for recording and retrieving birth information.
- **Birth Records:** Official information recorded at the time of birth.
- **Data Accuracy:** The correctness and completeness of recorded information.

CHAPTER 2

LITERATURE REVIEW

2.0 Introduction

The purpose of this chapter is to critically examine existing literature relevant to Digitalised Birth Record Management System. According to Creswell (2014), an effective literature review should justify the research problem, compare existing viewpoints, and demonstrate how the current study contributes new knowledge. The reviewed literature is assessed for relevance, credibility, and consistency, with emphasis placed on recent studies where available. Rather than merely describing previous work, this chapter compares and contrasts scholarly arguments in order to position the present study within the broader research debate.

2.1 Theoretical Literature Review

2.1.1 Birth Registration and Record Management Systems

The design and implementation of a digitalised birth record management system to replace handwritten registers. Literature consistently describes birth registration as a core component of health information systems and civil registration. UNICEF (2019) states that accurate birth records are essential for legal identity and access to social services. AbouZahr et al. (2015) agree that reliable birth data supports planning and evaluation of maternal and child health services.

However, several scholars contend that manual, paper-based systems undermine data quality and service efficiency. WHO (2018) notes that handwritten registers are prone to loss, physical damage, illegibility, and transcription errors. While these systems are inexpensive and easy to use, their limitations become pronounced as records accumulate over time. The literature therefore supports the need to replace physical systems with digital alternatives, directly justifying the first objective of this study.

2.1.2 Digital Health Record Systems

Digital health record systems are widely viewed as an improvement over physical record keeping. Laudon and Laudon (2020) state that digital systems enhance efficiency by improving storage, retrieval, and reporting of health data. Ozair et al. (2015) agree that electronic records reduce paperwork and improve accessibility of patient information.

Despite these advantages, the literature also highlights critical weaknesses. Katuu (2016) contends that many digital systems merely replicate physical processes in electronic form, thereby perpetuating errors associated with physical data entry. This suggests that digitisation alone is insufficient to guarantee accuracy and reliability. These findings support the study's second objective, which seeks to improve accuracy and speed of birth data management beyond basic digitisation.

2.2 Empirical Literature Review

Empirical studies provide practical evidence on the performance of digitalised health information systems. Fraser et al. (2017) found that electronic health record systems significantly reduced data retrieval time and improved reporting efficiency in hospital settings. Mutale et al. (2018) similarly reported improvements in data completeness and timeliness following the implementation of digital systems in district hospitals.

Studies focusing on digitalised systems report stronger outcomes. Zhou et al. (2019) observed that AI-supported validation mechanisms reduced data entry errors and improved record reliability. However, most empirical studies focus on general electronic health records rather than birth-specific record management systems. In the African and rural hospital context, empirical evidence on Digitalised Birth Record Management System remains limited, indicating a clear research gap.

2.3 Summary, Research Gap, and Significance

The reviewed literature demonstrates broad agreement that physical birth record systems are inefficient and prone to error, while digital systems offer improvements in efficiency but do not fully resolve accuracy challenges. The literature further contends that digitalised systems provide superior outcomes through automated validation, analytics, and decision support.

Research Gap

Despite extensive research on electronic health records studies in healthcare, there is limited empirical research focusing specifically on Digitalised Birth Record Management Systems in district hospitals within developing countries. This gap justifies the present study at Nyanga District Hospital.

Significance of the Study

The study is significant to healthcare workers by reducing administrative workload and improving record accuracy. Hospital management benefits from real-time dashboards that support planning and reporting. Academically, the study contributes to literature on AI applications in birth record management, an under-researched area in district hospital settings.

CHAPTER 3

FEASIBILITY STUDY

3.0 Introduction

A feasibility study assesses the workability of the proposed system in terms of cost, technology, operations, time, organizational alignment, and social acceptance.

The purpose is not to solve the problem at this stage, but to determine whether implementation is realistic and beneficial. The findings guide management in deciding whether to proceed with the project.

3.1 Economic Feasibility

The researcher examined the cost-benefit analysis of the project, which assesses whether it is possible to implement the project and determine whether the project can be attained within the firm's budget constraints.

3.1.1 Benefits of the proposed system

Tangible benefits

- Reduced administrative works
- Diminish wasted inventory.
- Reduced stationery costs through computerization.
- Improve cash flow.

Intangible benefits

- Reduced pitfalls.
- Increased employee motivation through reduced stress from less work.
- Improved data integrity and system reliability.

3.2 Technical Feasibility

The researcher assessed whether it is technically possible to carry out a project before taking up the project, plan and prepare for every step of the operation.

Nyanga District Hospital has basic ICT infrastructure, including electricity and computers. The proposed system requires standard hardware and commonly used database and software development tools.

3.3 Operational Feasibility

The researcher carried out the study to analyse and determine whether the needs of the organization can be met by completing the project. A user-friendly system is required so that it can be quickly accepted.

3.4 Organizational Feasibility

Organizational feasibility evaluates whether the system aligns with institutional goals.

The hospital aims to improve service delivery, data accuracy, and reporting efficiency. The proposed system supports these objectives by enhancing record management and enabling data-driven decision-making. Thus, the system is consistent with the hospital's strategic direction and is organizationally feasible.

3.5 Schedule Feasibility

The researcher calculated and continually re-examined whether it is possible to complete all amount and scope of work lying ahead, utilizing the given number of resources, within required period of time.

3.6 Project Plan

This is a roadmap for a digitalised system that efficiently capture, store and manage birth records accurately and securely.

3.6.1 Gantt Chart

Time period in:									
Months	**June**	**July**	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**	**Jan**	**Feb**
Proposal									
Feasibility study									
Requirement analysis									
Design									
Coding and testing									
Implementation and post implementation plan									
Summary, conclusion and Recommendations									

3.7 Conclusion

The feasibility analysis confirms that the proposed Digitalised Birth Record Management System is economically viable, technically achievable, operationally practical, organizationally aligned, schedule realistic, and socially acceptable.

CHAPTER 4

REQUIREMENTS ANALYSIS

4.0 Introduction

The researcher points out what users required in the proposed system and gave an insight of the operations of the current system, how processes are linked within the current system and how activities are going to be coordinated in the proposed system. The researcher used a set of gathering methodologies to collect data. The researcher analysed collected data highlighting the strengths and weaknesses of the current system and alternative solutions to the systems weaknesses.

4.1 Information gathering methodologies

4.1.1 Interviews

On two June two thousand and twenty-five the researcher asked questions to thirty staff members regarding their opinion on the efficiency and effectiveness of the current system and their desired changes. Interviews were carried in a space of one week, and all the staff members were interviewed in their offices. Notes were taken down during all interview sessions. The developer chose to interview midwifery, maternity nurses and administrative staff members separately so as to enable the gathering of all information from different levels and sources related to the line of birth registration and record, operations and activities undertaken by the by the hospital. A variety of questions were open-ended questions, closed-ended questions, and range of response questions. Interviews were documented and recorded during and after the process of interviewing.

Benefits

- Managed to clarify misconceptions about the current system
- Body expressions were notable during interviews, which made it easier to see which areas of the current system they were most uncomfortable with.
- The researcher obtained first-hand information.

Drawbacks

- Some of the interviewees failed to corporate and therefore did not give the interviewer enough time to gather all required information.
- Some questions were not answered correctly due to Ministry of Health law of confidentiality.

4.1.2 Questionnaires

On 9 June two thousand and twenty-five the researcher gave questionnaires to all staff members at the company. The purpose is to preview into the already existing system through the users' opinion. Questionnaires were to be returned in four days' time.

Benefits

- Questionnaires supply standard answers because the respondents were matching exactly to what the researcher needed to know.
- Questionnaires saved time.
- They were easy to arrange as compared to interviews.
- Since questions were the same it was easy to make conclusions.

Disadvantages

- Partially or poorly completed answers were provided as the respondent failed to understand some questions and also because there were no follow up questions that offered clarification.
- Pre-coded answers can be frustrating for the respondent thus deterring from answering them.

4.1.3 Observation

The researcher simply observed all activities that takes place at the company and managed to have the look and feel of what was actually transpiring throughout the firm.

Benefits

- Information obtained was from the actual site where problems were identified.

- Observations provided with a direct and fast source of information.
- The look and feel of current system enabled the information to be collected with less biased.

Disadvantages

- Findings are not accurate since the methodology used is based on assumptions based on human behaviour.
- As the members were aware that they were being observed, they likely changed their behaviour, which caused some of the data to be distorted.
- The method needs more personal commitment.

4.1 Analysis of the Current System

Birth records are physically recorded in handwritten registers in the maternity ward. Registers are physically stored and retrieved when parents request confirmation of birth. Monthly statistics are generated through physical counting.

- **Who records the data?** Nurses and maternity ward staff.
- **Who retrieves and issues records?** Health ICT Officers and Health Information Officers.
- **Problem:** Some parents delay collecting birth records for many years (e.g., births from 1997, 2004, 2013, 2016). Over time, registers become worn out, pages tear, ink fades, and records may be lost.

This leads to delays, administrative burden, and risk of missing data. Given the importance of birth documentation in Zimbabwe, this presents serious operational challenges.

4.2 Functional Requirements

The system shall:

1. Capture and store birth details electronically.
2. Validate entries to reduce errors.
3. Allow authorised users to search and retrieve records instantly.
4. Generate automated monthly and annual reports.

5. Provide a dashboard showing total births, live births, stillbirths, and trends.
6. Ensure secure login and data backup.

4.3 Non-Functional Requirements

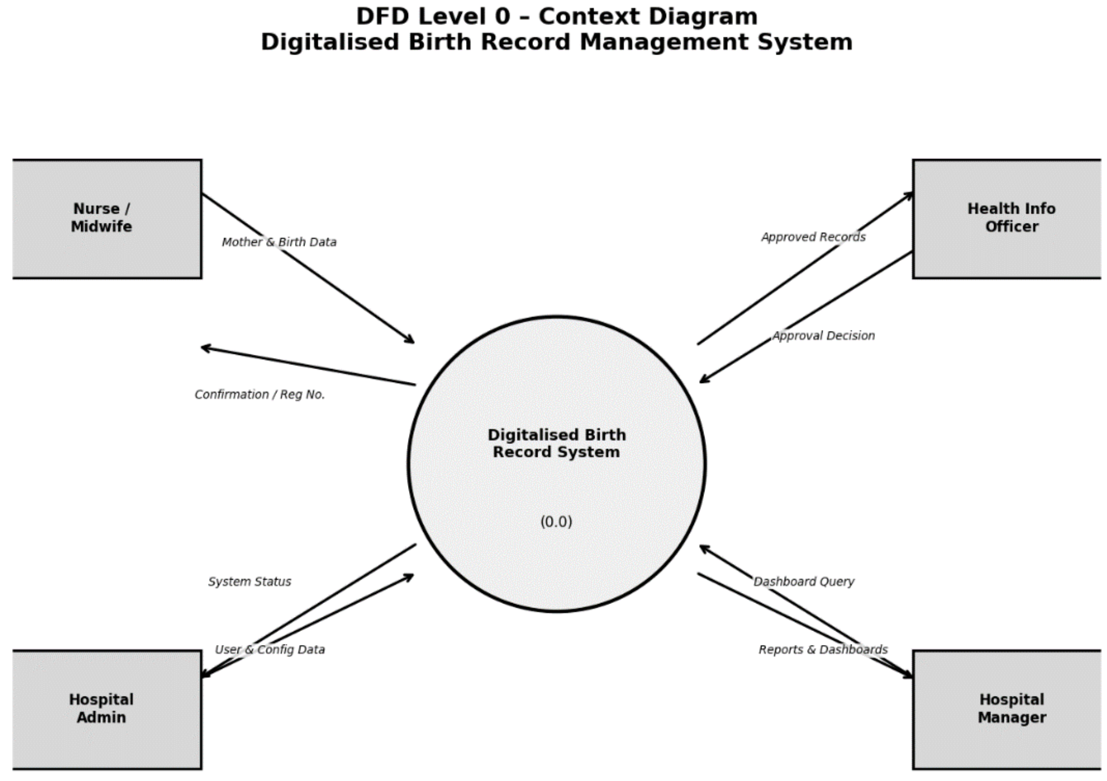
The system must:

- Be user-friendly and easy to learn.
- Operate on available hospital computers.
- Ensure data security and confidentiality.
- Function reliably with minimal downtime.
- Be cost-effective and sustainable within Zimbabwe's economic environment.

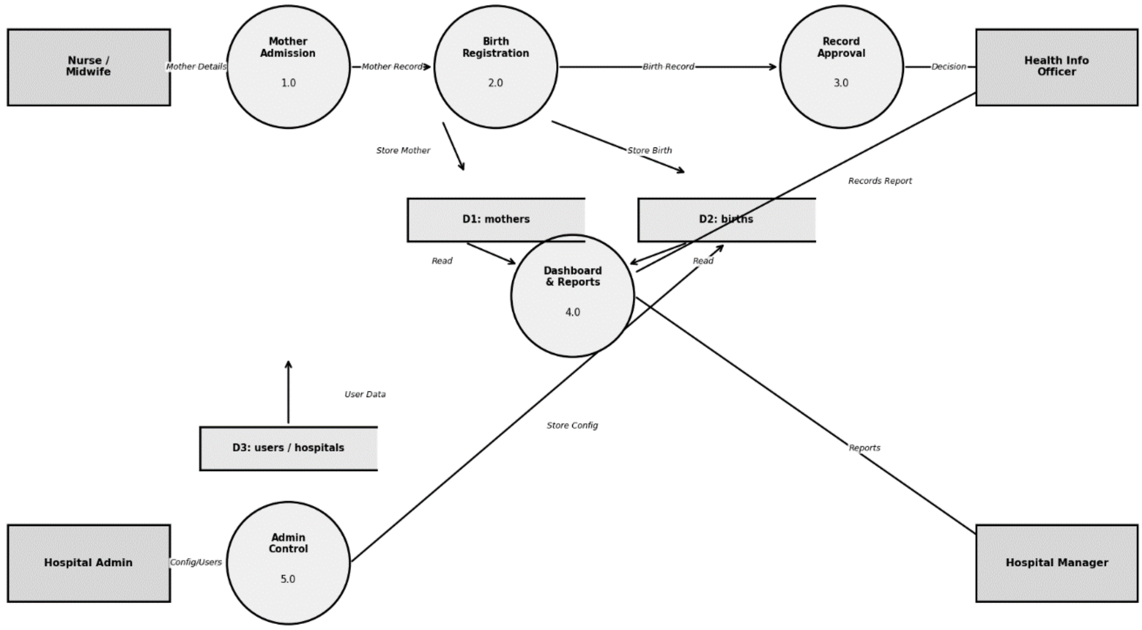
4.4 DATA ANALYSIS

The developer revealed processes and activities that are performed in the current system.

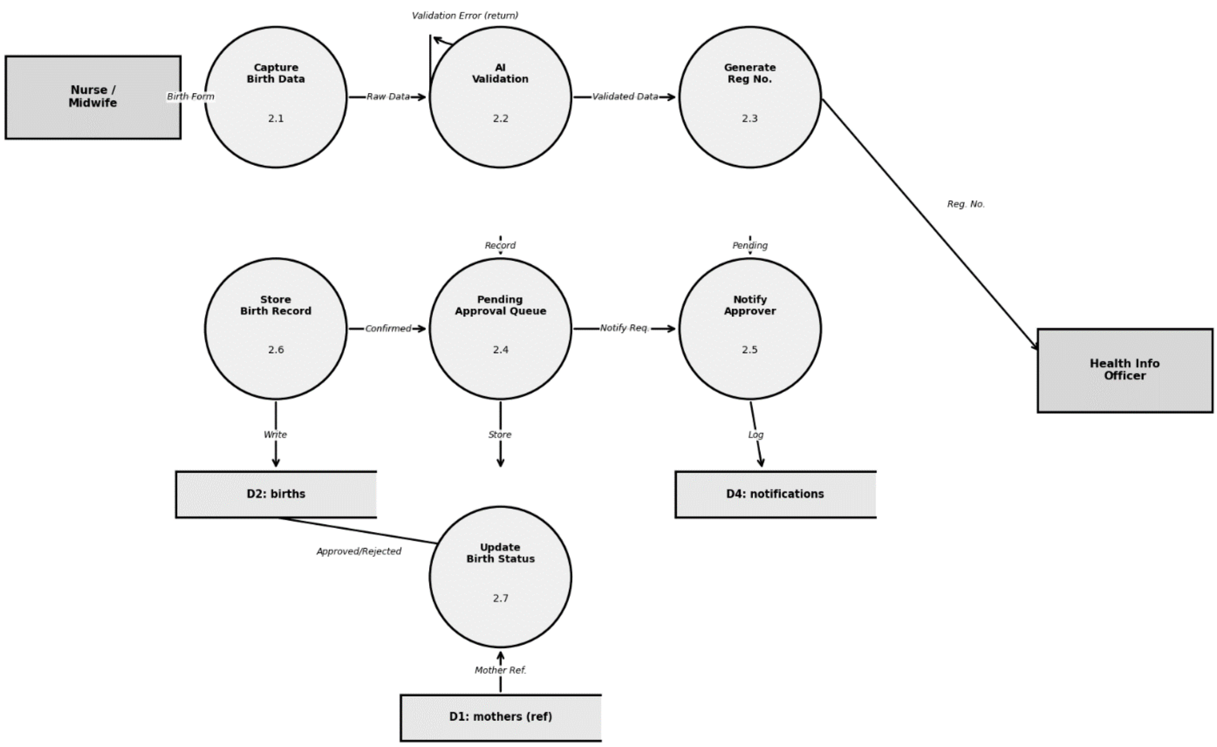
4.4.1 Data Flow Diagram



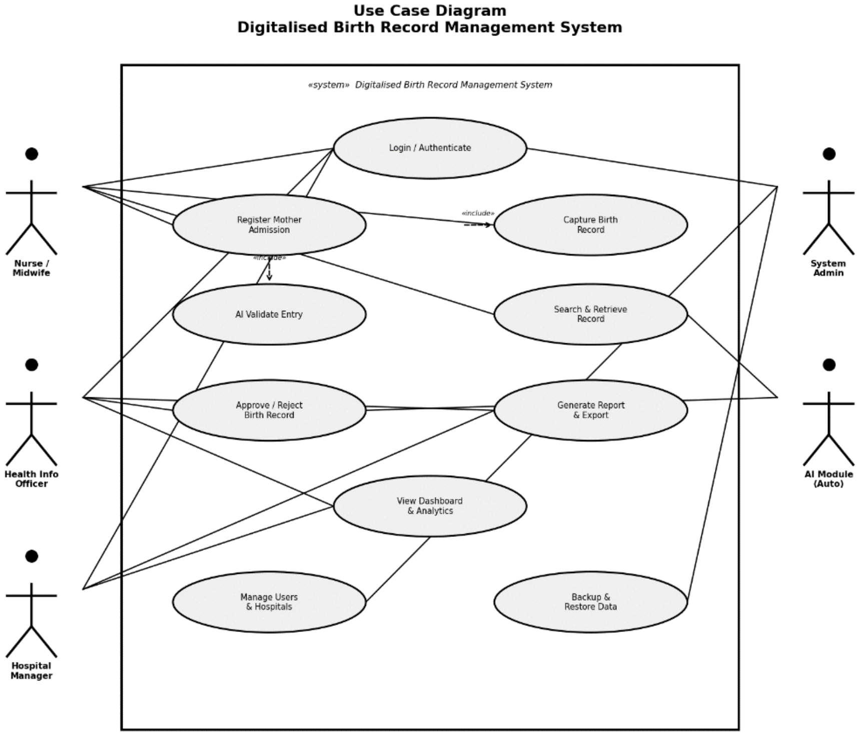
DFD Level 1 – Main System Processes
Digitalised Birth Record Management System



DFD Level 2 – Process 2.0: Birth Registration (Expanded)
Digitalised Birth Record Management System



4.4.2 Use Case



4.5 Conclusion

The current physical system is inefficient and vulnerable to data loss. The proposed digital system addresses these weaknesses through secure storage, instant retrieval, automated validation, and real-time reporting.

CHAPTER 5
SYSTEM DESIGN
5.0 Introduction

This chapter presents the design of the Digitalised Birth Record Management SystemRecord Management System for Nyanga District Hospital. The approved requirements from Chapter Four are transformed into a clear system design that explains how the system will operate.

The logical design developed during analysis is now converted into a physical design, detailing inputs, outputs, databases, processes, security controls, and system structure. This design serves as the blueprint for system development.

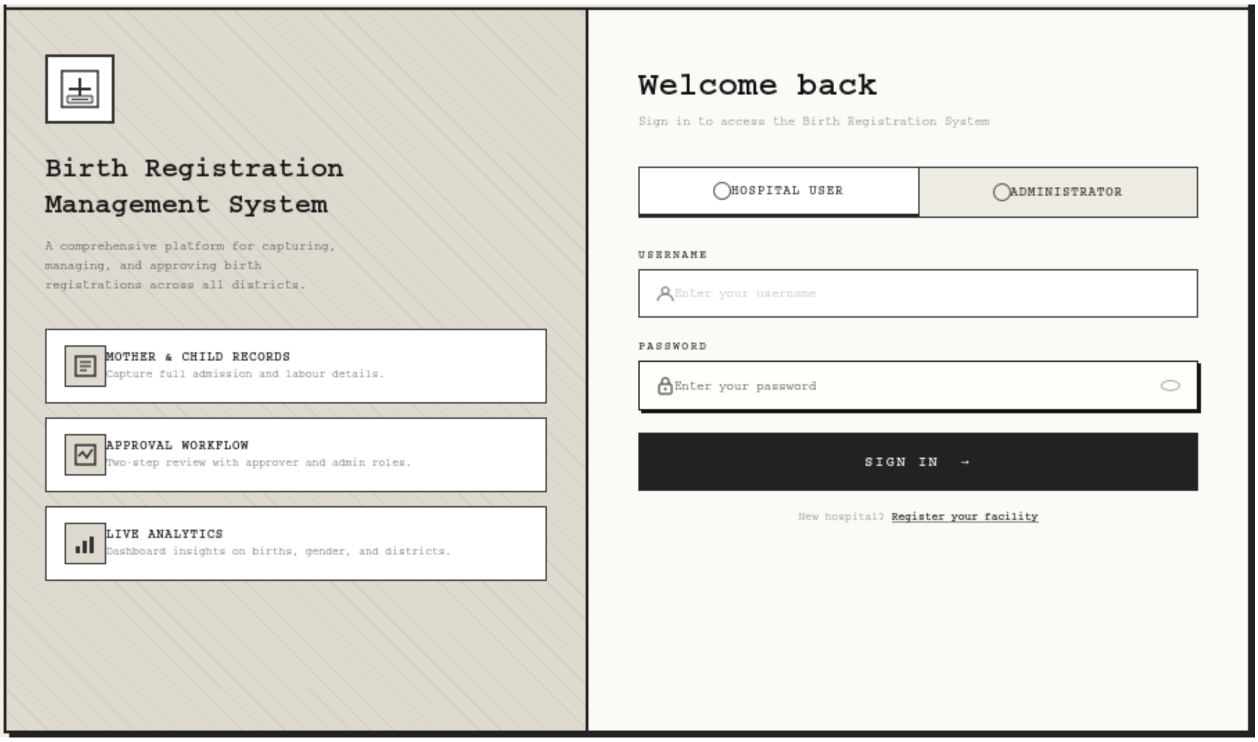
5.1 Interface Design

The system interface will be:

- Simple and user-friendly
- Menu-driven
- Designed for users with basic computer skills

Screens will include login pages, data entry forms, search functions, and dashboard views.

5.1.1 Interface Diagram



5.1 Database Tables

5.1.1 Admins table

Column Name	Data Type	Constraints	Description
Id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique admin identifier
Username	VARCHAR(100)	UNIQUE, NOT NULL	Admin login username
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
Email	VARCHAR(150)	UNIQUE, NOT NULL	Admin email address
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
last_login	DATETIME	NULLABLE	Last successful login time
is_active	TINYINT(1)	DEFAULT 1	1=Active, 0=Disabled account

5.1.2 Hospital Table

Column Name	Data Type	Constraints	Description
Id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique facility identifier
Name	VARCHAR(200)	NOT NULL	Official hospital name
Location	VARCHAR(200)	NOT NULL	Physical address / town
Province	VARCHAR(100)	NOT NULL	Province (e.g. Manicaland)
District	VARCHAR(100)	NOT NULL	District name
Phone	VARCHAR(20)	NULLABLE	Contact phone number
Email	VARCHAR(150)	NULLABLE	Official email address
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration timestamp
is_active	TINYINT(1)	DEFAULT 1	Facility active status

5.1.3 Users Table

Column Name	Data Type	Constraints	Description
Id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique staff user ID
hospital_id	INT	FK → hospitals(id), NOT NULL	Associated health facility
Username	VARCHAR(100)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
full_name	VARCHAR(200)	NOT NULL	Staff member full name
Role	ENUM	NOT NULL	'nurse','ict_officer','hio','manager'

Email	VARCHAR(150)	NULLABLE	Work email address
Phone	VARCHAR(20)	NULLABLE	Staff phone number
is_active	TINYINT(1)	DEFAULT 1	1=Active, 0=Disabled
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation time
last_login	DATETIME	NULLABLE	Last login timestamp

5.1.4 Mothers Table

Column Name	Data Type	Constraints	Description
Id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique mother admission ID
hospital_id	INT	FK → hospitals(id), NOT NULL	Admitting hospital
reg_number	VARCHAR(50)	UNIQUE, NOT NULL	Auto-generated registration number
full_name	VARCHAR(200)	NOT NULL	Mother's full legal name
date_of_birth	DATE	NOT NULL	Mother's date of birth
national_id	VARCHAR(30)	NULLABLE	National Identity Card number
Address	TEXT	NULLABLE	Residential address
Phone	VARCHAR(20)	NULLABLE	Contact phone number
marital_status	ENUM	NULLABLE	'single','married','widowed','divorced'
blood_group	VARCHAR(5)	NULLABLE	Blood group (e.g. O+)
admission_date	DATE	NOT NULL	Date admitted to maternity ward
Ward	VARCHAR(50)	NULLABLE	Ward assigned to
created_by	INT	FK → users(id), NOT NULL	Nurse who captured the admission
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	DATETIME	ON UPDATE CURRENT_TIMESTAMP	Last update timestamp

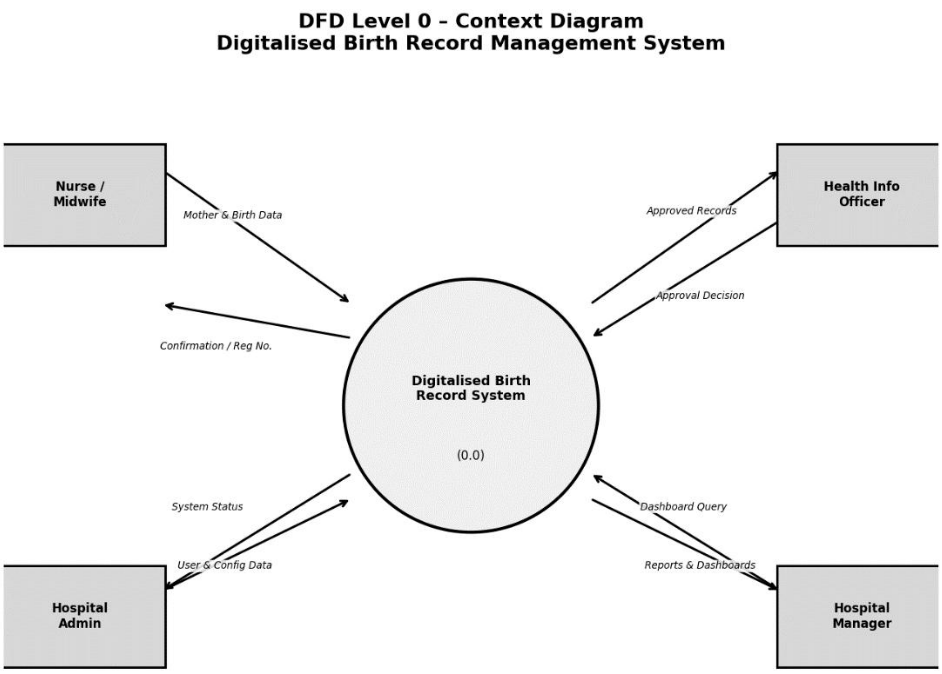
5.1.5 Birth Table

Column Name	Data Type	Constraints	Description
Id	INT	PK, AUTO_INCREMENT, NOT NULL	Unique birth record ID
mother_id	INT	FK → mothers(id), NOT NULL	Mother linked to birth
registered_by	INT	FK → users(id), NOT NULL	Nurse who captured the birth
approved_by	INT	FK → users(id), NULLABLE	ICT Officer who approved record
birth_date	DATE	NOT NULL	Date of birth of newborn

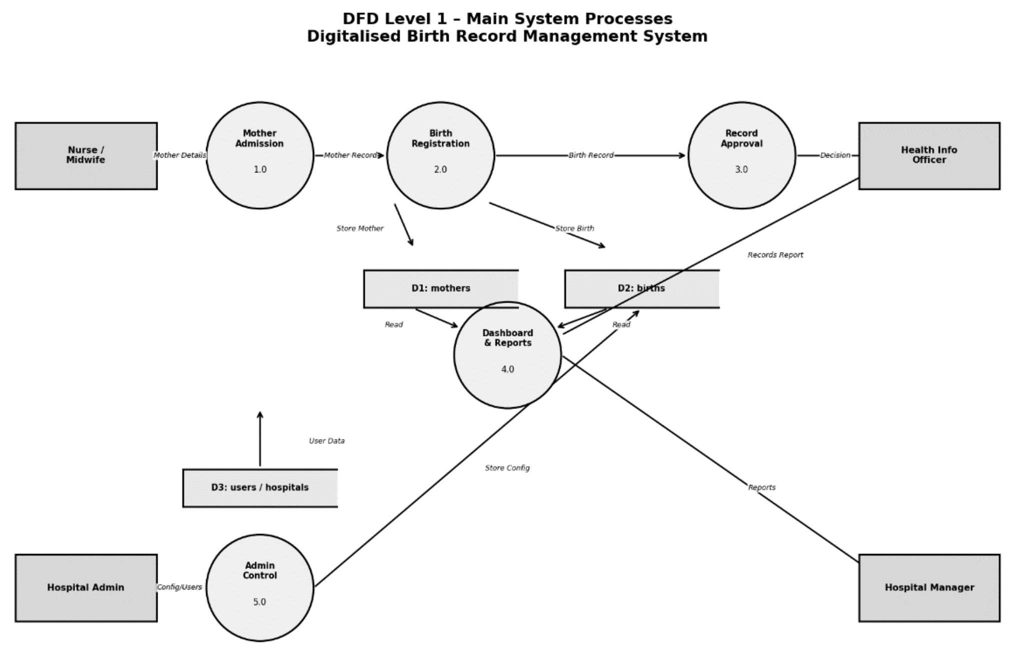
birth_time	TIME	NOT NULL	Time of birth
newborn_name	VARCHAR (200)	NULLABLE	Newborn name (if given at admission)
Sex	ENUM	NOT NULL	'male','female','indeterminate'
delivery_type	ENUM	NOT NULL	'normal','caesarean','assisted'
Outcome	ENUM	NOT NULL	'live_birth','stillbirth','neonatal_death'
gestational_age	INT	NULLABLE	Gestational age in weeks
weight_kg	DECIMAL(4,2)	NOT NULL	Birth weight in kilograms
apgar_score	TINYINT	NULLABLE	APGAR score at birth (0–10)
Complications	TEXT	NULLABLE	Notes on delivery complications
Status	ENUM	DEFAULT 'draft'	'draft','pending','approved','rejected','issued'
ai_flags	JSON	NULLABLE	AI validation flags and error messages
rejection_reason	TEXT	NULLABLE	Reason for record rejection
approved_at	DATETIME	NULLABLE	Approval timestamp
issued_at	DATETIME	NULLABLE	Certificate issue timestamp
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp
updated_at	DATETIME	ON UPDATE CURRENT_TIMESTAMP	Last update timestamp

5.2 Data Flow Diagrams (DFDs)

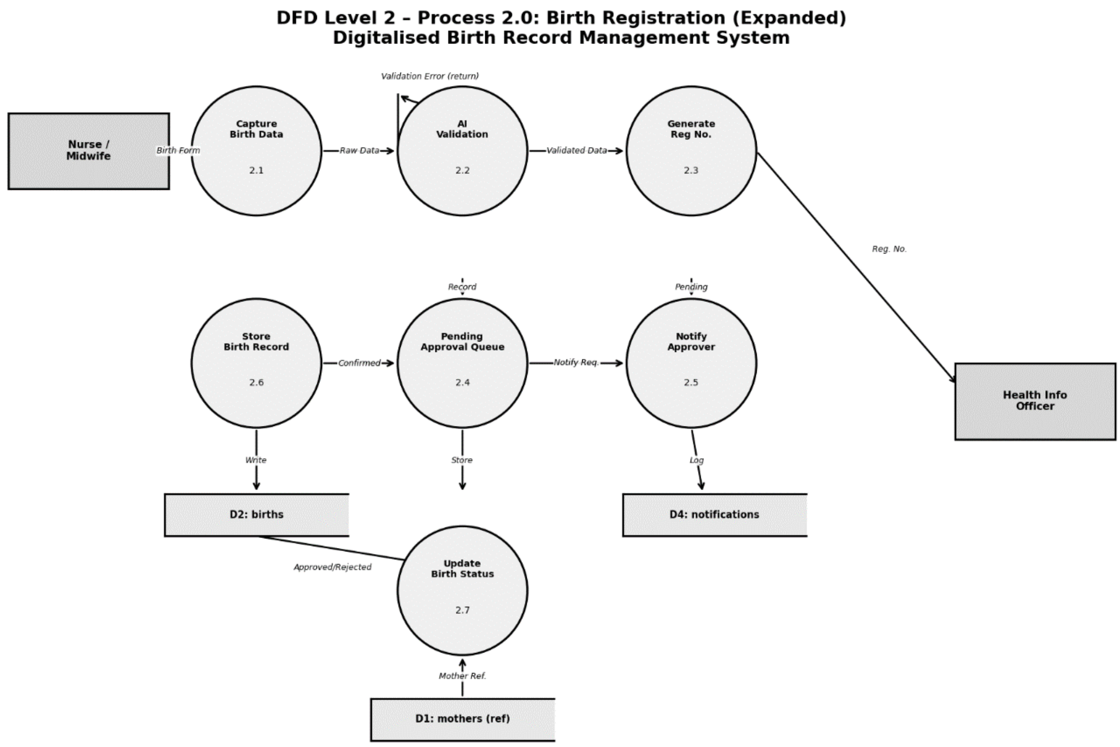
5.2.1



5.2.2


DFD Level 1 – Main System Processes
Digitalised Birth Record Management System
Nurse / Midwife
Mother Details
Mother Admission
1.0
Mother Record
Birth Registration
2.0
Birth Record
Record Approval
3.0
Decision
Health Info Officer
Store Mother
Store Birth
Records Report
D1: mothers
D2: births
Read
Read
Dashboard & Reports
4.0
User Data
Store Config
Reports
D3: users / hospitals
Hospital Admin
Config/Users
Admin Control
5.0
Hospital Manager

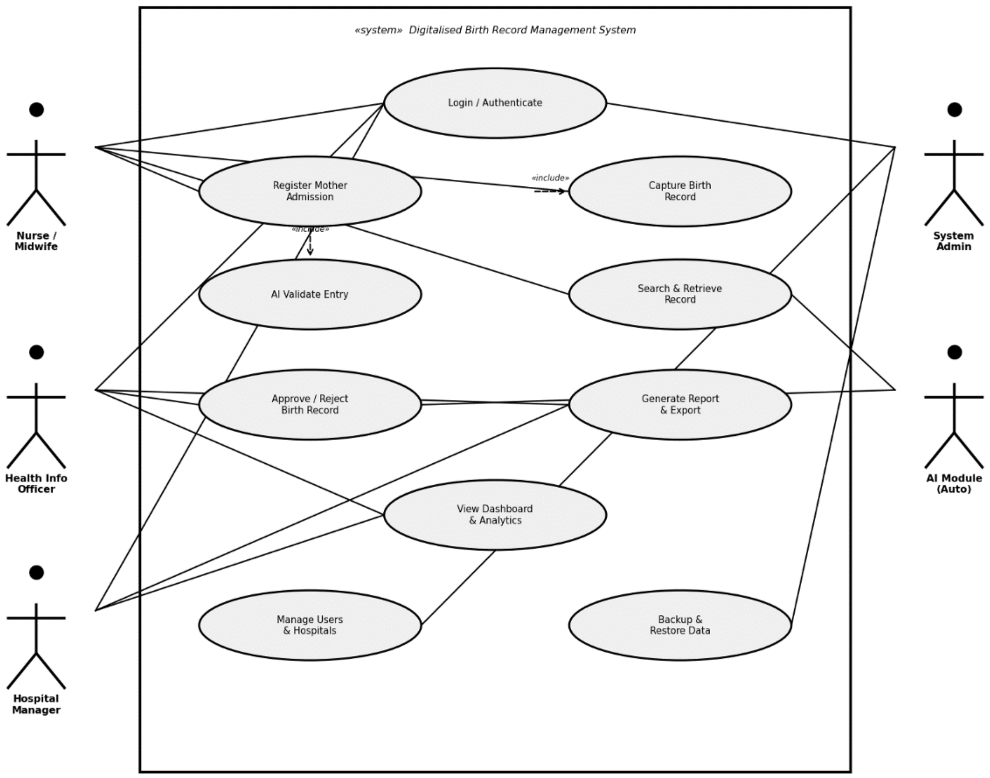
5.2.3


DFD Level 2 – Process 2.0: Birth Registration (Expanded)
Digitalised Birth Record Management System
Validation Error (return)
Nurse / Midwife
Birth Form
Capture Birth Data
2.1
Raw Data
AI Validation
2.2
Validated Data
Generate Reg No.
2.3
Reg. No.
Record
Pending
Store Birth Record
2.6
Confirmed
Pending Approval Queue
2.4
Notify Req.
Notify Approver
2.5
Health Info Officer
Write
Store
Log
D2: births
D4: notifications
Approved/Rejected
Update Birth Status
2.7
Mother Ref.
D1: mothers (ref)

5.3 UML Diagrams

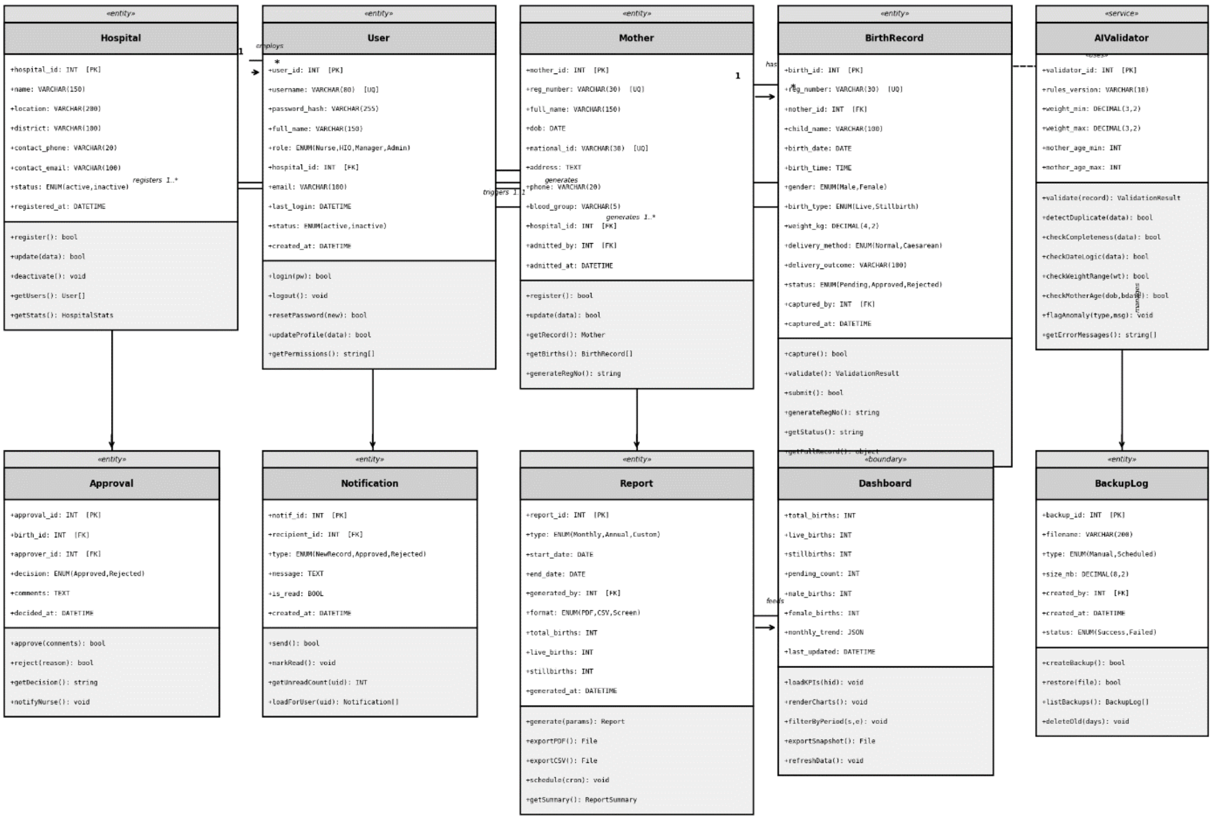
5.3.1 Use Case Diagram

Digitalised Birth Record Management System

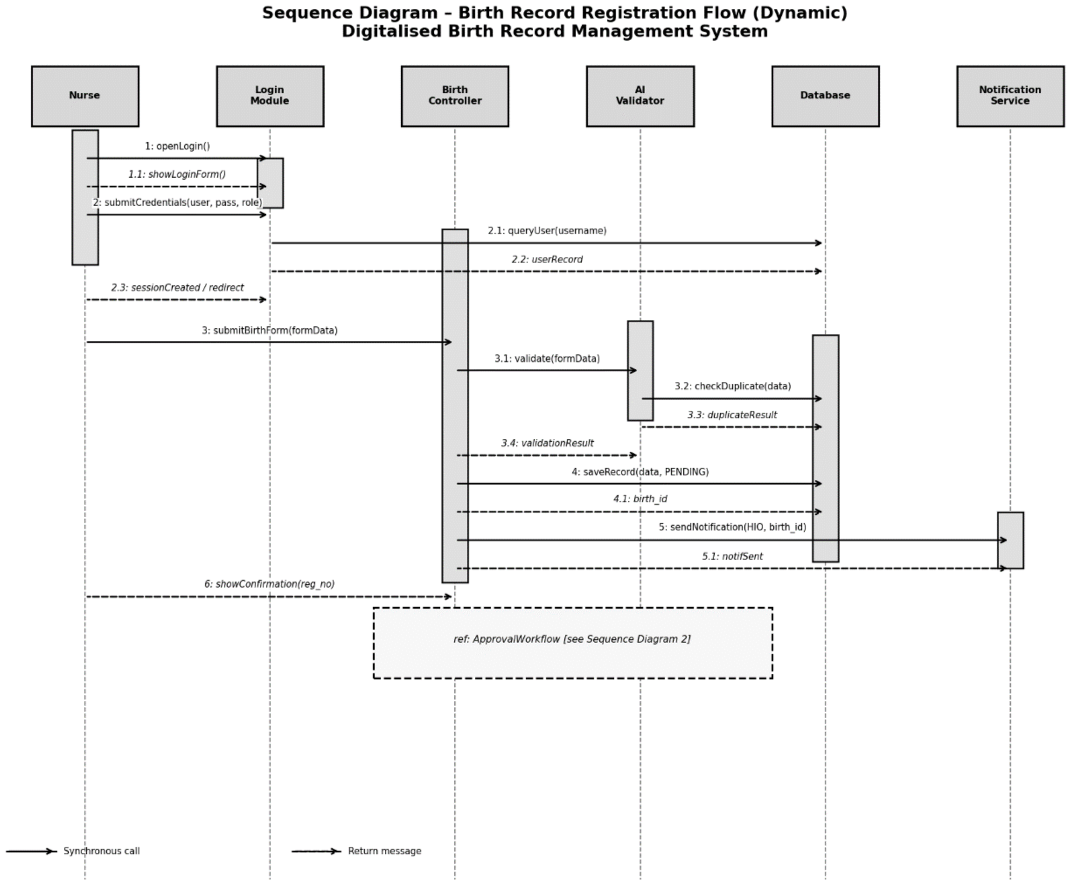


5.3.2 Class Diagram

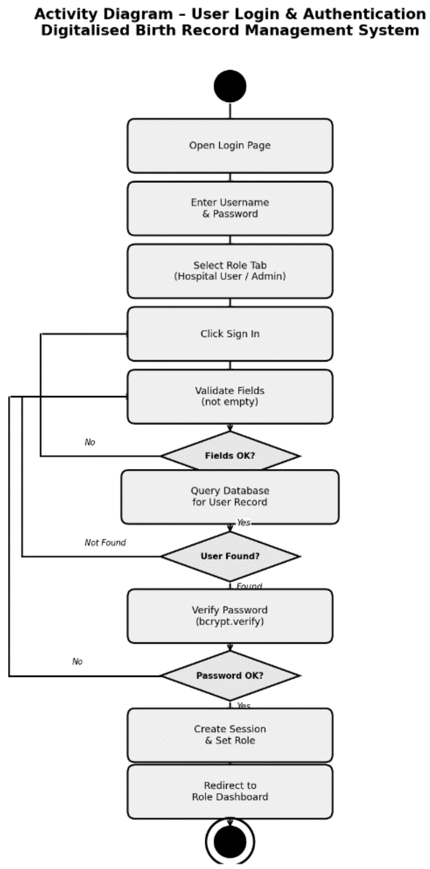
Digitalised Birth Record Management System

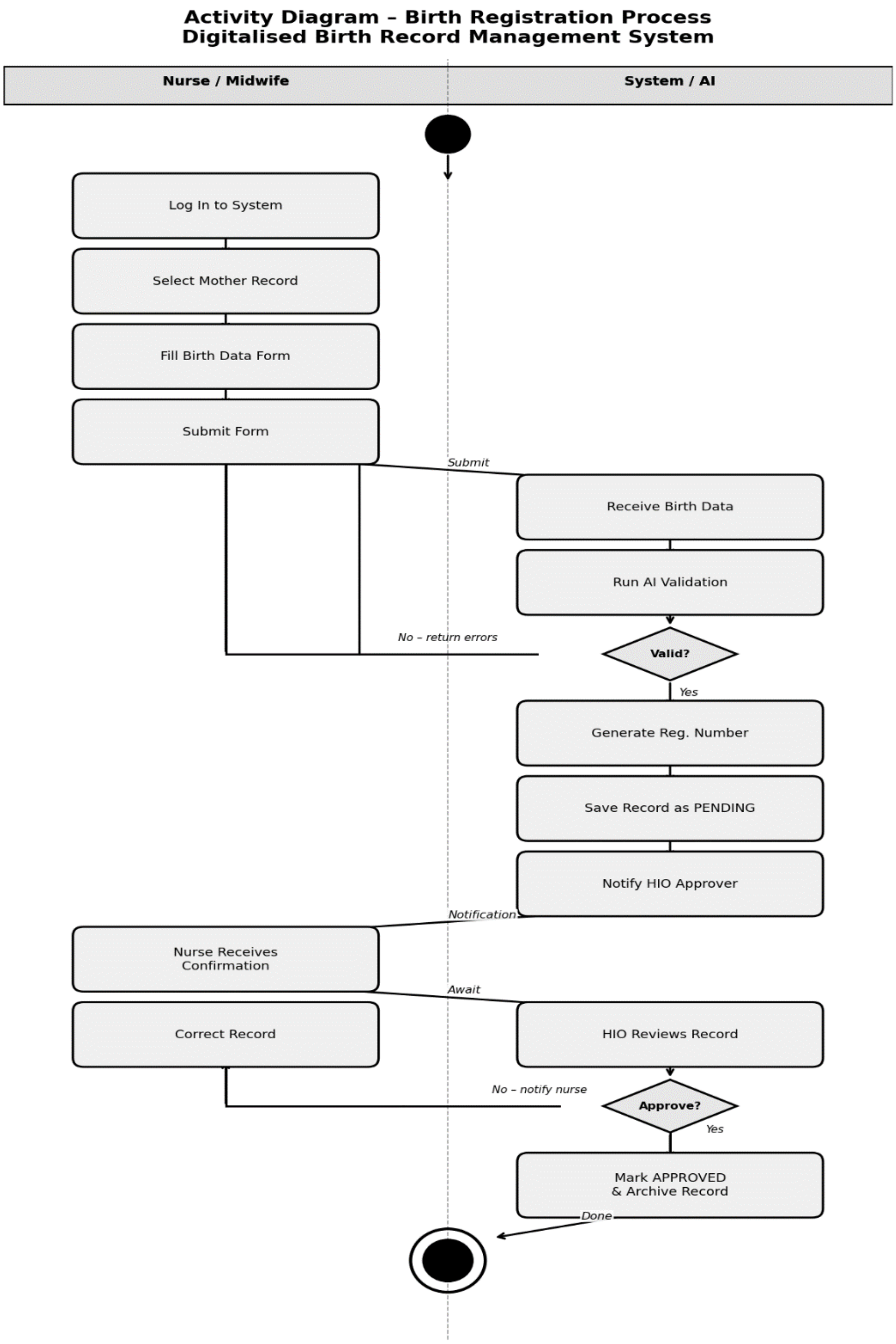


5.3.3 Sequence Diagram

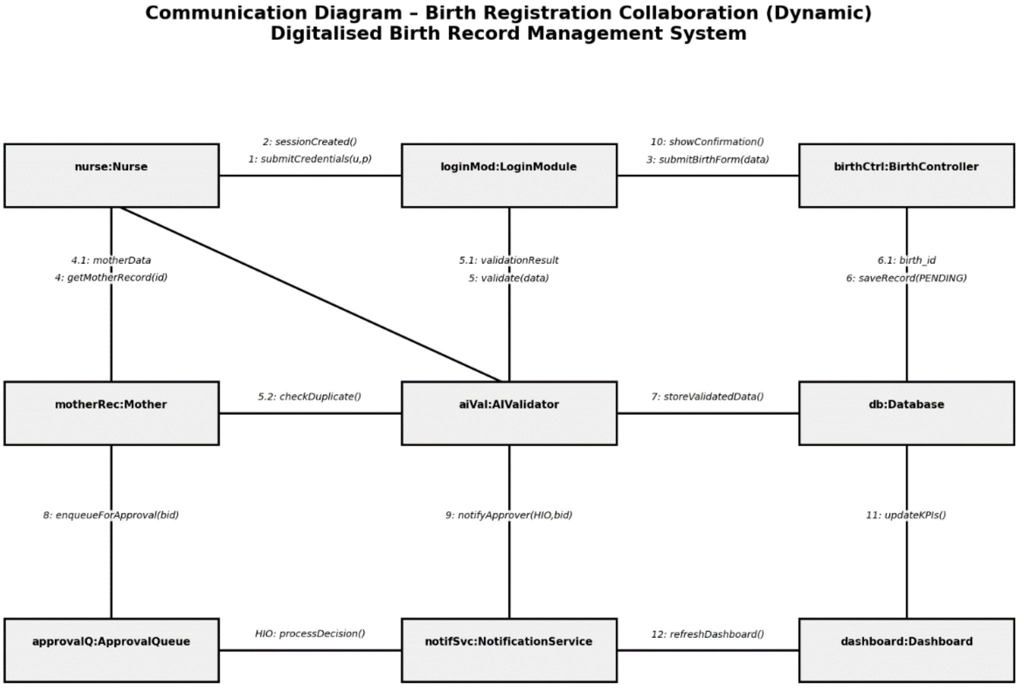


5.3.5 Activity diagram

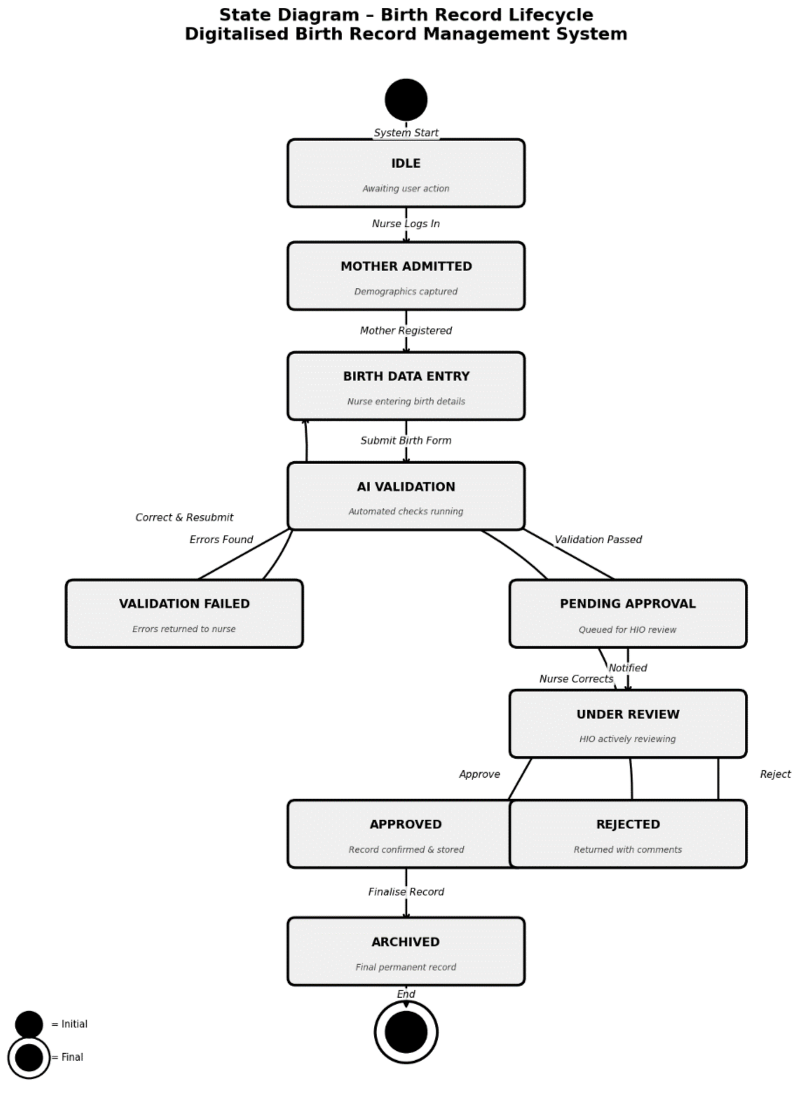



Activity Diagram – Birth Registration Process
Digitalised Birth Record Management System
Nurse / Midwife
System / AI
Log In to System
Select Mother Record
Fill Birth Data Form
Submit Form
Submit
Receive Birth Data
Run AI Validation
No – return errors
Valid?
Yes
Generate Reg. Number
Save Record as PENDING
Notify HIO Approver
Notification
Nurse Receives Confirmation
Await
Correct Record
HIO Reviews Record
No – notify nurse
Approve?
Yes
Mark APPROVED & Archive Record
Done

5.3.5 Communication Diagram



5.3.5State Diagram



5.5 Database Design (Entity Structure)

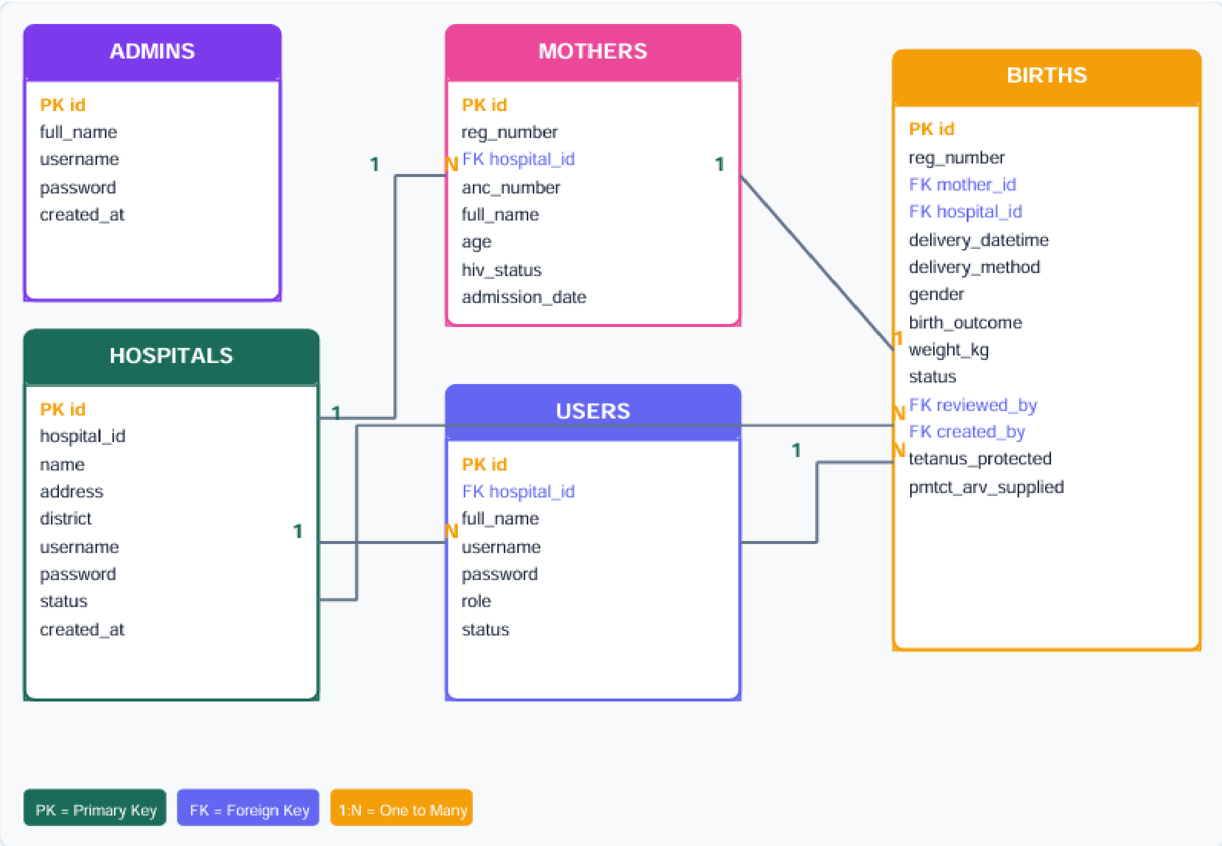
The system will use a structured database with key entities such as:

- Mother
- Newborn
- Birth Record
- User (Health ICT Officer, Nurse, Administrator)

Each record will have unique identifiers to prevent duplication. A data dictionary will define each data field to ensure consistency and accuracy.

5.5.1 Entity Relationship Diagram

The database contains five core tables. hospitals is the top-level entity representing registered facilities. Each hospital has many users (staff) and many mothers (admissions). Each mother can have one or more births records. The admins table is independent and manages the super-administrator accounts.



Tables	Primary Key	Key Relationships	Purpose
admins	id	-	Super administrator accounts
hospitals	id	1→N users, 1→N mothers	Registered health facilities
Users	id	N→1 hospitals	Hospital staff accounts
Mothers	id	N→1 hospitals, 1→N births	Maternal admission records
Births	id	N→1 mothers, N→1 users (x2)	Birth registration records

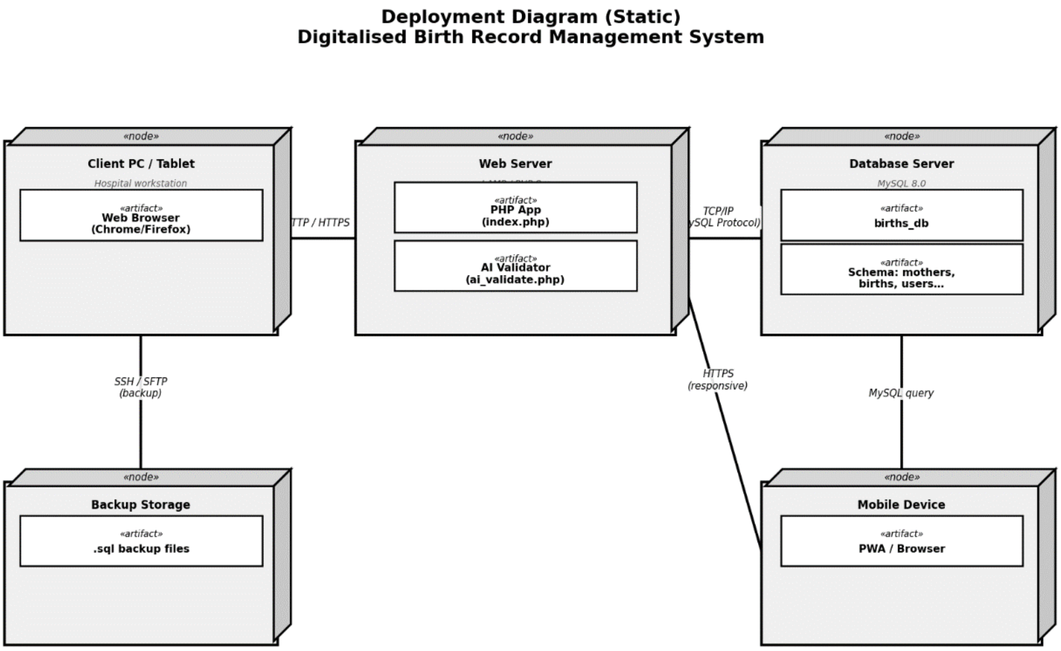
5.3 System Architecture Design

The proposed system will operate as a computer-based application within the hospital network. It will consist of:

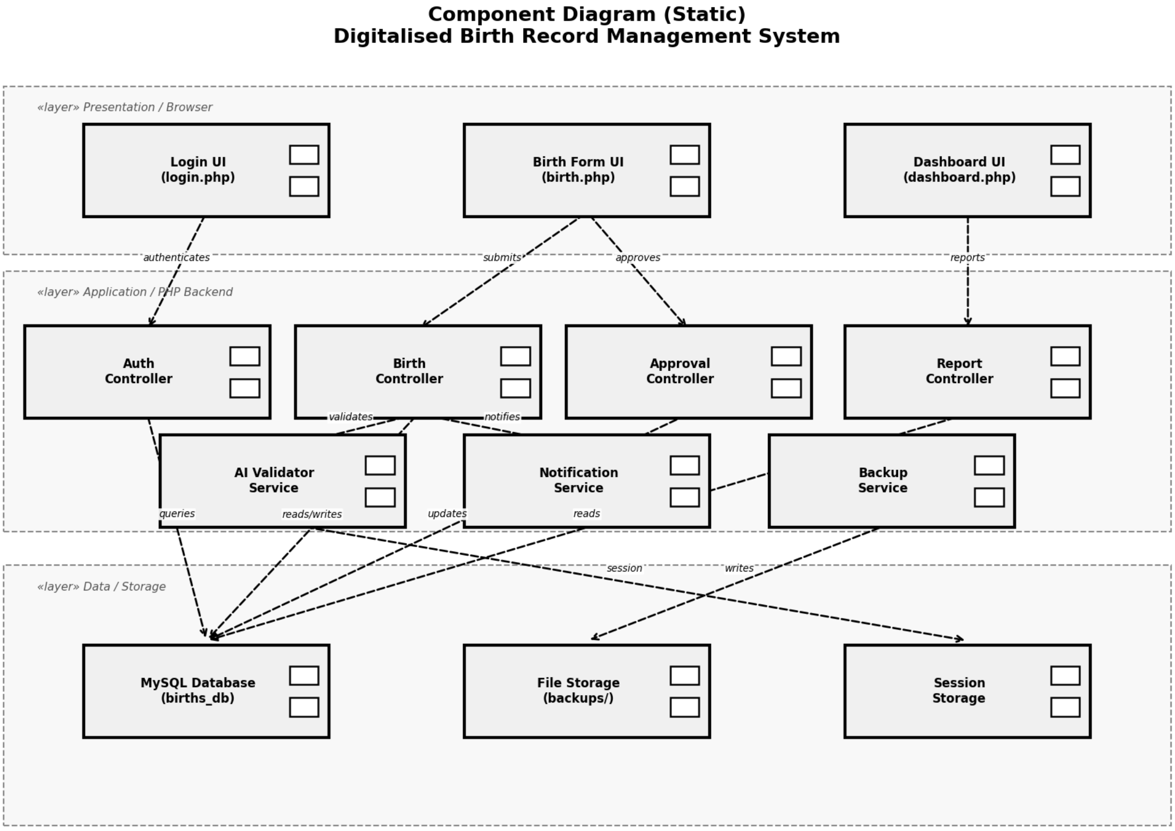
- A user interface for data entry and retrieval
- A central database for storing birth records
- A reporting and dashboard module
- An AI validation module for error detection

The system will run on standard hospital computers and will not require expensive infrastructure, ensuring suitability within Zimbabwe's economic environment.

5.3.1 Deployment Diagram



5.1.2 Component Diagram



5.2 Input Design

Inputs refer to the data captured into the system. These include:

- Mother's details
- Newborn details
- Date and time of birth
- Type of delivery
- Birth outcome (live birth or stillbirth)

Digital forms will mirror the current physical registers to ensure familiarity for users. Mandatory fields and automated validation rules will reduce incomplete or incorrect entries.

5.3 Process Design

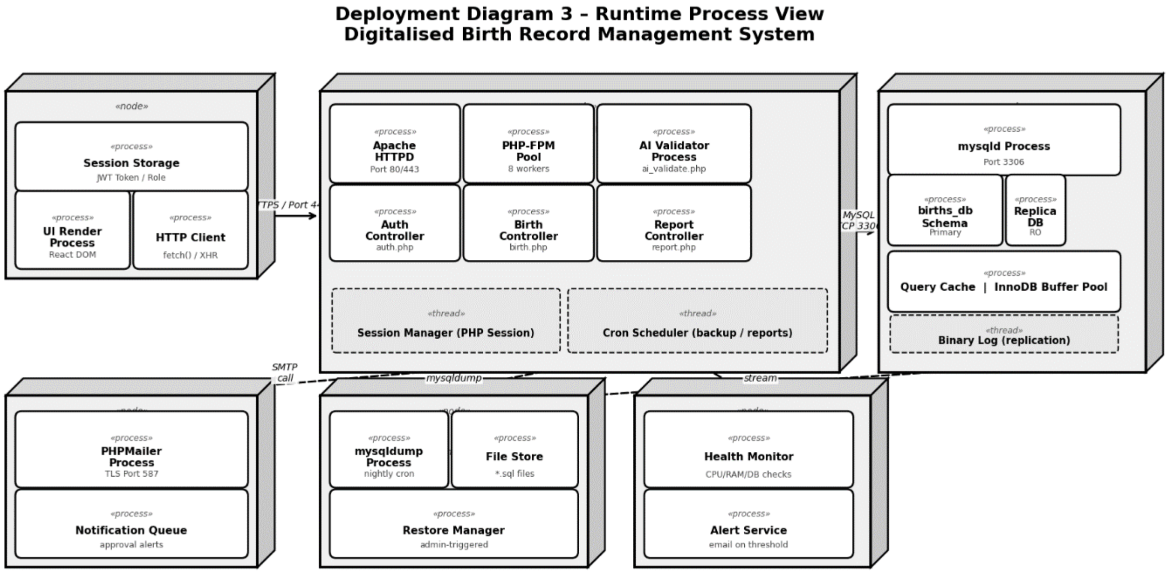
The system will perform the following processes:

1. Capture and validate birth data.
2. Store records securely in the database.

3. Allow authorised users to search and retrieve records.
4. Generate automated reports and statistical summaries.
5. Display real-time trends on a dashboard.

AI functionality will detect missing data, duplicate entries, and inconsistencies during data entry.

5.3.1 Deployment Diagram



5.3.1 User Flow Flowchart

Two parallel flows govern the system. The Hospital User flow (left) covers login, mother admission, birth capture, and the approval pipeline. The Admin flow (right) covers hospital registration approval, user creation, and system monitoring. Diamond shapes represent decision points; rounded rectangles are processes.

5.4 Output Design

The system outputs will include:

- Individual birth record retrieval
- Monthly and annual statistical reports
- Dashboard summaries showing:
    - Total births
    - Live births
    - Stillbirths
    - Delivery trends

Outputs will be clear, printable, and suitable for management reporting and decision-making.

5.6 Security and Control Design

Security is critical due to the sensitive nature of birth records. The system will include:

- Secure login authentication
- Role-based access control
- Data encryption
- Regular data backups
- Audit trails to track user activity

Security risks will be assessed and managed before deployment.

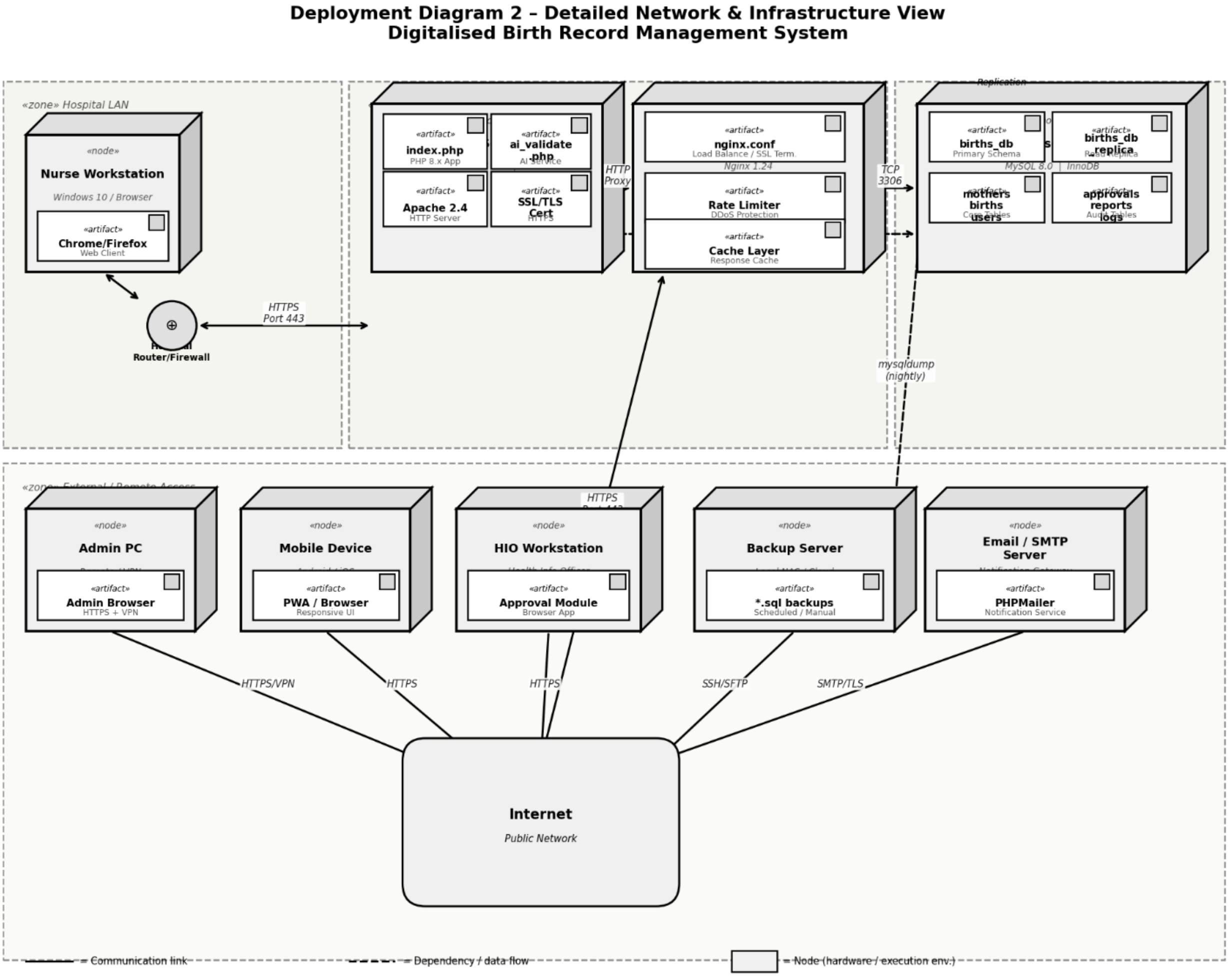
5.8 Hardware and Software Platform

The system will operate on:

- Standard desktop computers or laptops
- A relational database management system
- A secure local server or structured storage system

The selected platform will be cost-effective, sustainable, and compatible with existing hospital resources.

5.8.1 Deployment Diagram



5.9 Operational Design Considerations

The design also addresses:

- User training requirements
- Backup procedures
- System documentation
- Maintenance and technical support
- Staffing roles and responsibilities

These considerations ensure smooth implementation and long-term sustainability.

5.10 Conclusion

This chapter transformed the approved requirements into a structured system design. The logical design has been converted into a physical blueprint covering inputs, processes, outputs, database structure, security, and platform requirements.

The design ensures that the proposed system is secure, practical, and aligned with the operational realities of Nyanga District Hospital. Upon approval, the project can proceed to the Development Phase.

CHAPTER 6

CODING AND TESTING

6.0 Introduction

This chapter describes the implementation of the system design through coding and testing. At this stage, the approved system design is translated into a working software application. The objective is to develop a reliable, error-free system that meets the specified requirements.

6.1 Coding (Programming Phase)

Coding involves converting the system design specifications into computer programs using an appropriate programming language. The selected language and development tools were chosen based on system requirements, compatibility, cost-effectiveness, and sustainability within the hospital environment.

The system was developed using a modular approach. Each module (data entry, validation, database management, reporting, and dashboard) was programmed separately to ensure:

- Easier development and debugging
- Improved maintenance
- Flexibility for future upgrades

Compilers and development tools were used to detect and remove syntax errors. Logical errors, which affect how the system behaves, were identified through careful testing and corrected accordingly.

Well-structured and documented code was maintained to reduce future maintenance challenges.

6.2 Testing

Testing ensures that the developed system functions according to the specified requirements. It verifies accuracy, reliability, and performance before full deployment.

A structured test plan was developed and executed using prepared test data and later actual data.

6.3 Program Testing

Each module of the system was tested individually. This included:

- Verifying data entry validation rules
- Checking database storage accuracy
- Testing report generation functions
- Confirming correct dashboard calculations

Errors identified during this stage were corrected through debugging. The objective was to ensure each program functioned correctly on its own.

6.4 System Testing

After individual modules were tested, they were integrated and tested as a complete system.

System testing involved:.

- Running the system using actual hospital data
- Verifying that outputs matched expected results
- Testing interaction between modules
- Checking security controls and user access levels

If discrepancies were identified, corrections were made and retesting was conducted until the system operated as expected.

6.5 User Acceptance Testing

Once technical testing confirmed system stability, users (nurses, Health ICT Officers, and Health Information Officers) were invited to test the system using real operational data.

This ensured that:

- The system met user expectations
- Workflows were properly supported

- Outputs were accurate and acceptable

Feedback from users was considered before final deployment.

6.6 Conclusion

The coding and testing phase transformed the system design into a functional and reliable application. Through modular development, structured program testing, system integration testing, and user acceptance testing, errors were identified and corrected.

The system was confirmed to operate according to specified requirements and is ready for implementation.

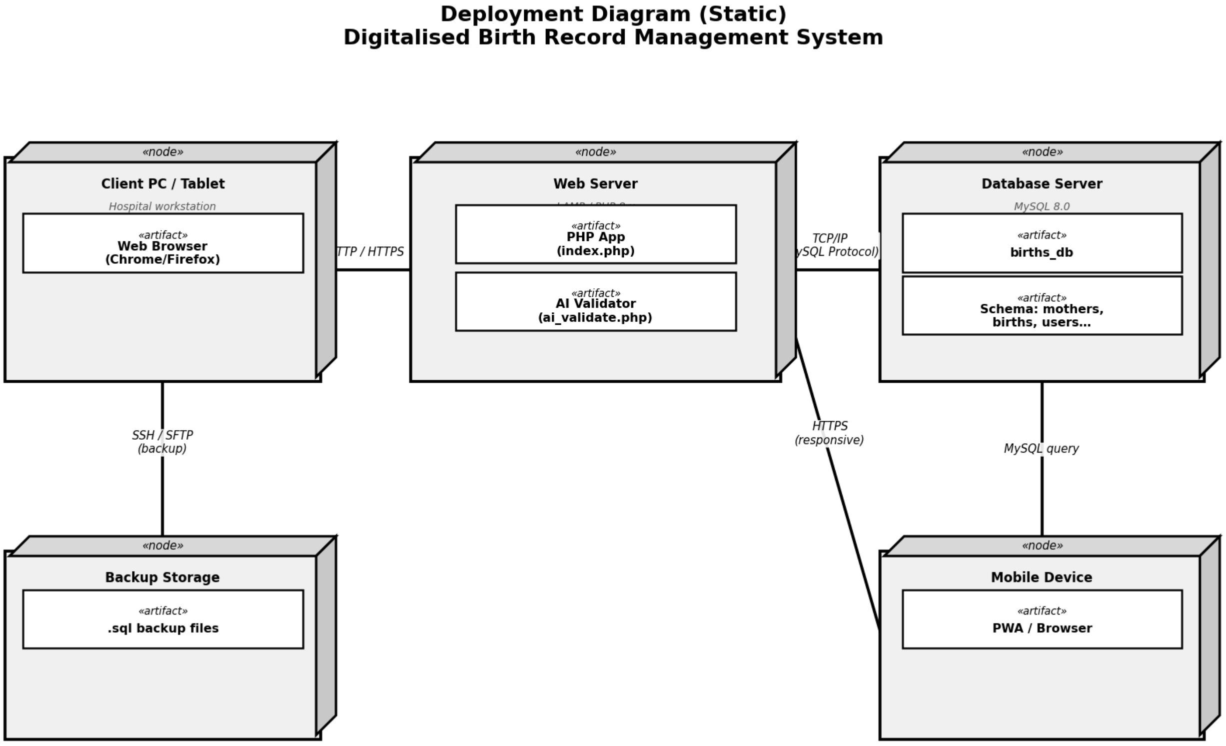
CHAPTER 7

IMPLEMENTATION AND POST-IMPLEMENTATION PLAN

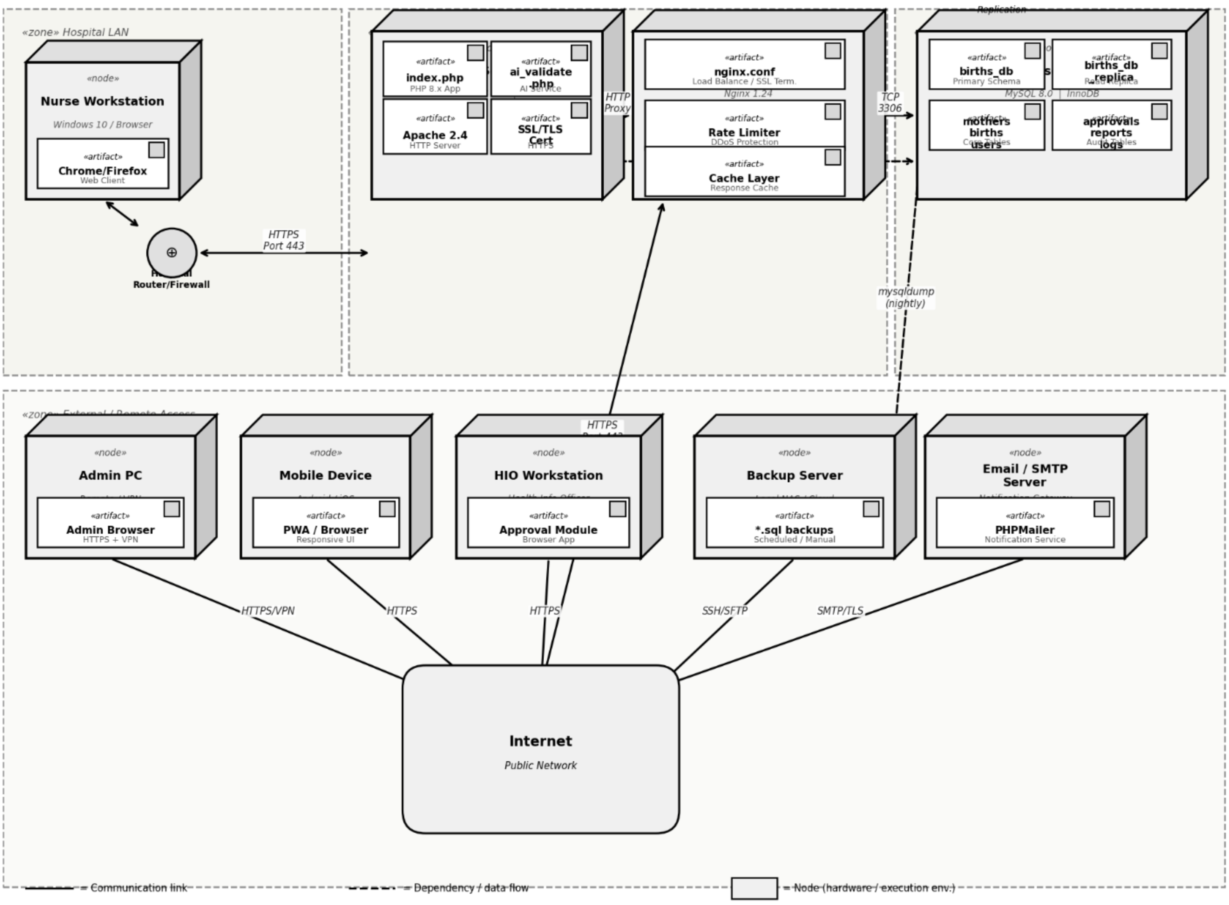
7.0 Introduction

This chapter describes the deployment of the Digitalised Birth Record Management System and the activities required after installation. Implementation converts the developed system into practical use within Nyanga District Hospital.

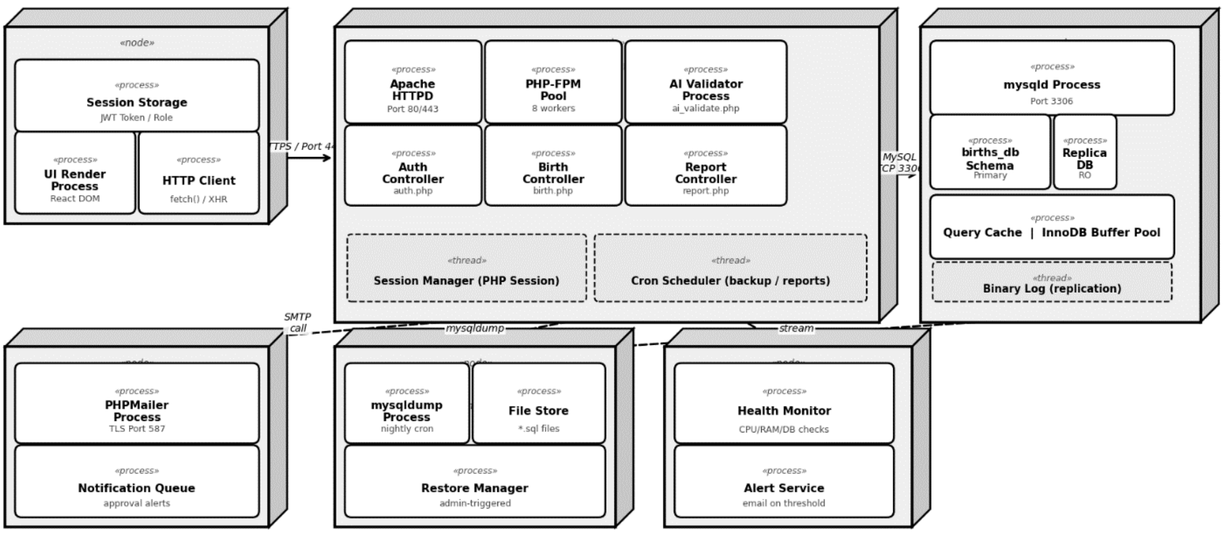
7.1.1 Deployment Diagrams (1 -3)



Deployment Diagram 2 – Detailed Network & Infrastructure View
Digitalised Birth Record Management System



Deployment Diagram 3 – Runtime Process View
Digitalised Birth Record Management System



7.2 System Implementation

After user acceptance testing, the system is prepared for operational use. The key steps include:

7.2.1 Acquisition and Installation

- Installation of required hardware (computers, storage devices).
- Setup of the database with defined security and backup procedures.
- Installation of the application software on user machines.

The system is configured to operate within existing hospital infrastructure to ensure cost-effectiveness and sustainability.

7.2.2 Data Conversion

Relevant data from the physical registers is entered into the new system where necessary. Proper validation is conducted to ensure accuracy during migration.

Data security, access control, and recovery procedures are fully defined before full operation.

7.2.3 User Training

Training is conducted for nurses, Health ICT Officers, and Health Information Officers. Training focuses on:

- Logging into the system
- Capturing birth data
- Searching and retrieving records
- Generating reports and dashboard summaries

The aim is to ensure confidence and smooth transition from physical to digital processes.

7.3 Changeover Strategy

To minimise operational risk, a Parallel Run Strategy is recommended.

Under this approach:

- Both physical and digital systems operate simultaneously for a defined period.
- Outputs from both systems are compared.
- Errors are corrected before full migration.

This approach reduces risk while maintaining service continuity.

7.4 Documentation

Proper documentation ensures system continuity and sustainability.

7.4.1 User Documentation

Provides clear instructions on:

- System operation
- Data entry procedures
- Report generation
- Common error messages

7.4.2 System Documentation

Includes:

- System design details
- Database structure
- Program specifications
- Data dictionary
- Process descriptions

This supports future upgrades and maintenance.

7.5 Maintenance and Post-Implementation Plan

Maintenance ensures the system continues to operate efficiently. It includes:

- Correcting identified errors
- Monitoring system performance
- Regular data backups

- Updating security measures
- Improving features where necessary

Periodic system reviews will assess:

- Performance efficiency
- User satisfaction
- Need for enhancements

If major changes are required, a new development cycle may be initiated.

7.6 Conclusion

This chapter outlined the steps for system deployment, user training, changeover, documentation, and maintenance. Proper implementation and ongoing monitoring ensure that the system remains reliable, secure, and aligned with hospital needs.

With successful implementation, the hospital transitions from a vulnerable physical system to a secure, efficient digital birth record management system.

CHAPTER 8

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

8.0 Introduction

This chapter presents a summary of the study, the major findings, conclusions drawn from the research, and recommendations for future action and further research.

8.1 Summary of the Study

This dissertation investigated the development of the proposed system for Nyanga District Hospital. The aim was to replace the manual, paper-based birth registration system with a secure, efficient, and intelligent digital system.

The study:

- Examined weaknesses in the current physical system.
- Reviewed literature on digital health records and AI in healthcare.
- Conducted feasibility and requirements analysis.
- Designed, developed, tested, and implemented a proposed system.

The objectives were achieved through literature review, system analysis, design, coding, and system testing.

8.2 Summary of Major Findings

This study has shown that the current physical birth registration system is inefficient, vulnerable to data loss, and time-consuming.

One of the most significant findings to emerge from this study is that delayed collection of birth records (for children born many years earlier) increases the risk of missing, damaged, or incomplete records.

The study also found that:

- Physical reporting increases administrative workload.
- Retrieval of historical records is slow and unreliable.

- Lack of digital storage limits accurate statistical reporting.

The results of this investigation show that a digital system with automated validation improves data accuracy, speeds up retrieval, and enhances reporting efficiency.

These findings suggest that integrating digitalised system and dashboard reporting significantly strengthens birth record management in a district hospital setting.

8.3 Significance of the Findings

The findings contribute practically and academically.

Practically, the study provides a cost-effective and sustainable digital solution suitable for the Zimbabwean healthcare environment. It addresses real operational challenges faced by nurses, Health ICT Officers, and Health Information Officers.

Academically, the study contributes to knowledge on AI applications in birth record management, an area with limited research in district hospitals within developing countries.

The study confirms that digital transformation in health information systems improves efficiency, data integrity, and decision-making.

8.4 Conclusions

Based on the findings, the following conclusions are drawn:

- The physical birth registration system is inefficient and vulnerable to long-term data deterioration.
- The proposed system effectively addresses these weaknesses.
- The system improves accuracy, retrieval speed, and reporting capabilities.
- The research hypothesis that the digital system improves accuracy is supported.

The study confirms that transitioning to a secure digital system is both necessary and feasible for Nyanga District Hospital.

8.5 Recommendations

8.5.1 Recommendations from the Research Findings

This research has brought about the following recommendations:

- The hospital should adopt and fully implement the proposed digital birth record management system.
- Physical registers should be gradually phased out after a parallel run period.
- Regular system maintenance and data backups should be enforced.

8.5.2 Recommendations for Further Research

It is recommended that further research be undertaken in the following areas:

- Integration of the system with national civil registration systems.
- Expansion of the system to other hospital departments.
- Evaluation of long-term impact of AI validation on data quality.

8.5.3 Recommendations for Practice and Policy

The findings of this study have important implications for health policy and hospital management.

- There is a definite need for district hospitals to prioritise digital record management systems.
- Policymakers should support cost-effective digital health solutions suitable for resource-limited environments.
- Training in digital health information management should be strengthened for healthcare staff.

8.6 Final Statement

This study demonstrates that a secure, digitalised system can significantly improve birth record management at Nyanga District Hospital. The transition from physical to digital processes is essential for improving efficiency, safeguarding data, and supporting informed healthcare planning.